



Khulna University of Engineering & Technology, Khulna
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OPEN ENDED LAB Project

Title :TTL-Based Aerospace Cabin Safety Logic Controller

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TTL-Based Aerospace Cabin Safety Logic Controller

Abstract

TTL-Based Aerospace Cabin Safety Logic Controller is a digital electronics project designed using basic 7400-series TTL logic gates to simulate aircraft cabin safety monitoring operations. The system monitors different cabin conditions through input switches representing pilot emergency status, passenger seat condition, and lavatory occupancy. The controller processes these inputs using combinational logic circuits and provides visual indications through three LEDs. The green LED represents a safe operating condition, the yellow LED indicates a warning state, and the red LED represents an emergency or precautionary condition. The project is implemented completely using hardware-based logic gates without any microcontroller or programmable device. This project demonstrates the practical application of Boolean algebra, combinational logic design, and TTL digital electronics in aerospace-inspired safety monitoring systems.

1.Objectives

The primary objectives of this project are as follows:

1. To design and implement an aerospace cabin safety controller using TTL logic gates.
2. To understand the practical application of combinational logic circuits in safety monitoring systems.
3. To implement Boolean logic functions using 7400-series TTL integrated circuits.
4. To simulate different aircraft cabin operational conditions using switches and LED indicators.
5. To analyze system outputs using truth tables and logic gate operations.
6. Develop knowledge of digital electronics and hardware-based safety control systems.

2.Introduction

Safety monitoring is an essential part of modern aerospace systems because aircraft cabins must remain secure and stable during different flight conditions. Various electronic monitoring systems are used in aircraft to detect abnormal situations and provide warning or emergency indications whenever necessary. The *TTL-Based Aerospace Cabin Safety Logic Controller* is developed to demonstrate a simple aircraft cabin monitoring system using basic digital logic gates from the 7400-series TTL family. The project applies combinational logic principles to analyze different cabin conditions and produce suitable output indications. In this system, input switches are used to represent different operational conditions such as pilot emergency activation, passenger seat status, and lavatory occupancy. The outputs are displayed using LEDs that indicate normal, warning, and emergency states of the aircraft cabin. The controller is designed completely using hardware logic gates including AND, OR, NOT, and NAND gates. The project helps in understanding digital logic implementation, Boolean operations, and practical applications of TTL circuits in aerospace-inspired safety control systems.

3. Background Theory

3.1 TTL Logic Gates

TTL (Transistor-Transistor Logic) is a widely used digital logic family in electronic circuit design. TTL integrated circuits operate using binary voltage levels where HIGH represents logic '1' and LOW represents logic '0'. The 7400-series TTL ICs are commonly used for implementing digital logic operations because of their reliability, fast switching speed, and simple hardware implementation. In this project, basic TTL logic gates such as AND, OR, and NOT gates are used to implement the aerospace cabin safety controller. These logic gates process different aircraft cabin conditions and generate suitable output indications according to Boolean logic operations.

3.2 AND Gate

The AND gate produces a HIGH output only when all its inputs are HIGH. If any input becomes LOW, the output also becomes LOW. AND gates are commonly used in digital systems where multiple conditions must be satisfied simultaneously before generating an output signal. In this project, the AND gate is used to detect critical combined aircraft conditions where multiple warning states occur at the same time.

3.3 OR Gate

The OR gate produces a HIGH output when at least one input is HIGH. The output becomes LOW only when all inputs are LOW. OR gates are widely used in monitoring and warning systems where any abnormal condition can activate an alert signal.

In this project, the OR gate is used to activate precautionary or emergency indications whenever one or more warning conditions become active.

3.4 NOT Gate

The NOT gate, also known as an inverter, produces the opposite logic level of the input signal. When the input is HIGH, the output becomes LOW. When the input is LOW, the output becomes HIGH.

In this project, the NOT gate is used to create complementary logic conditions required for generating safe operational states and proper output control.

3.5 Combinational Logic Circuit

A combinational logic circuit is a digital circuit whose output depends only on the present input conditions. These circuits do not contain memory elements or feedback paths. Output is determined directly from Boolean logic expressions and truth table analysis.

The *TTL-Based Aerospace Cabin Safety Logic Controller* is implemented as a combinational logic circuit where different aircraft cabin input conditions are processed through TTL logic gates to generate safe, warning, and emergency indications using LEDs.

4.Circuit Design and Schematic

4.1:Circuit Schematic

The figure below shows the complete Proteus simulation schematic of the *TTL-Based Aerospace Cabin Safety Logic Controller*. The circuit is designed using basic 7400-series TTL logic gates along with input switches and LED indicators. The switches represent different aircraft cabin operational conditions, while the LEDs indicate safe, warning, and emergency states according to the implemented combinational logic circuit.

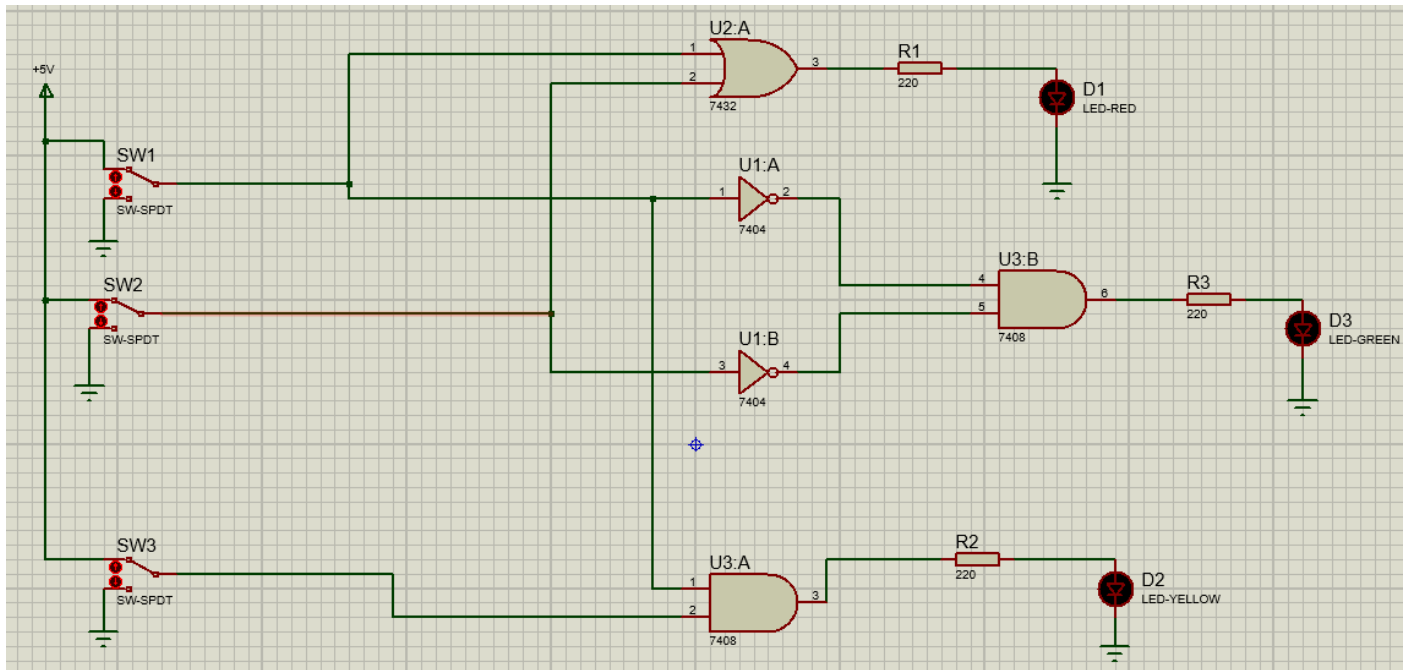


Figure-1:Proteus simulation schematic of the *TTL-Based Aerospace Cabin Safety Logic Controller*

4.2:Component List

Ref.	Components
U1:A,U1:B	7404 NOT gate
U2:A	7432 OR gate
U3:A,U3:B	7408 AND gate
R1	Resistor(220 Ohm)
R2	Resistor(220 Ohm)
R3	Resistor(220 Ohm)
LED	Green,Yellow,Red
DC Power Supply	5V

5.System Operation

5.1 Input Monitoring Subsystem

The input section of the circuit consists of three switches representing different aircraft cabin operational conditions: Pilot Master Emergency Switch (P), Seat Unbuckled/Vacated Switch (S), and Lavatory Door Lock Switch (B). These switches generate binary logic inputs that are processed by the TTL combinational logic circuit. The Pilot Master Emergency Switch represents emergency or precautionary conditions initiated from the flight deck. The Seat Unbuckled/Vacated Switch indicates passenger instability or seatbelt warning conditions. The Lavatory Door Lock Switch represents lavatory occupancy status during flight operation. The input conditions are continuously monitored by the logic circuit to determine the current operational state of the aircraft cabin.

5.2 Logic Processing Subsystem

The logic processing section is implemented using basic 7400-series TTL gates including AND, OR, and NOT gates. These gates process the input conditions according to predefined Boolean logic expressions and truth table analysis. The OR gate activates the precautionary or emergency indicator whenever pilot emergency activation or passenger warning conditions occur. The AND gate detects combined hazardous conditions where multiple critical inputs are active simultaneously. The NOT gate is used to generate complementary logic conditions required for safe-state indication. The outputs generated by the logic gates depend only on the present input conditions, making the system a combinational logic circuit.

5.3 Output Indication Subsystem

The output section consists of three LEDs representing different aircraft operational states. The green LED indicates a normal and safe cabin condition when no warning or emergency inputs are active. The red LED indicates precautionary or emergency conditions requiring passenger attention and safety measures. The yellow LED indicates critical hazard conditions where multiple warning states occur simultaneously. The LED outputs provide a simple visual indication of aircraft cabin status according to the logic implemented in the controller circuit.

Table 1: Truth table of the TTL-Based Aerospace Cabin Safety Logic Controller

P	S	B	Aircraft Operational condition	Red LED(F)	Yellow LED(R)	Green LED(C)
0	0	0	Normal cabin condition	OFF	OFF	ON
0	0	1	Restroom occupied during normal operation	OFF	OFF	ON
0	1	0	Passenger unbuckled warning	ON	OFF	OFF
0	1	1	Passenger unbuckled and restroom occupied	ON	OFF	OFF
1	0	0	Pilot emergency/precaution active	ON	OFF	OFF
1	0	1	Emergency condition with restroom occupied	ON	ON	OFF
1	1	0	Emergency with passenger instability	ON	OFF	OFF
1	1	1	Maximum precautionary/emergency condition	ON	ON	OFF

6. Discussion

6.1 Simulation Results

The *TTL-Based Aerospace Cabin Safety Logic Controller* was successfully designed and simulated using basic 7400-series TTL logic gates (NOT, OR, AND). The circuit operated according to the designed truth table and produced the expected output indications for different aircraft cabin operational conditions. The input switches representing pilot emergency status, passenger seat condition, and lavatory occupancy were tested using different binary combinations. The logic gates correctly processed these inputs and activated the corresponding LEDs based on the implemented combinational logic design. During normal operating conditions, the green LED remained ON to indicate a safe cabin state. When passenger warning or pilot precautionary conditions became active, the red LED turned ON successfully. Under combined hazardous conditions, the yellow LED also activated correctly to indicate a critical warning state. The simulation confirmed that the TTL logic gates performed the required Boolean operations accurately and generated stable outputs without noticeable errors. The project successfully demonstrated the practical implementation of digital logic design in an aerospace-inspired cabin safety monitoring application.

7. Limitations

Despite successful simulation and operation, the *TTL-Based Aerospace Cabin Safety Logic Controller* has several limitations:

7.1. Limited Input Conditions

The system monitors only three predefined aircraft cabin conditions. Real aircraft safety systems require a much larger number of sensors and monitoring parameters such as smoke detection, temperature monitoring, pressure control, and fire alarm systems.

7.2. No Automatic Sensor Integration

The project uses manual switches to simulate aircraft conditions instead of real electronic sensors. Therefore, the system cannot automatically detect real-time cabin changes or environmental conditions.

7.3. Hardware-Based Fixed Logic

Logic operations are implemented permanently using TTL gates. Any modification in system behavior or operational conditions requires redesigning the hardware connections and logic circuit.

7.4. No Data Storage or Feedback System

The controller operates only as a combinational logic circuit and does not store previous states or provide feedback analysis. Output depends only on current input conditions.

7.5. Limited Scalability

As the number of monitoring conditions increases, the circuit complexity also increases significantly. Additional safety features would require more logic gates, wiring, and hardware components.

7.6. Simulation-Based Implementation

The project was primarily designed and tested in simulation and small-scale hardware implementation. It is not suitable for direct real-world aerospace applications without advanced safety certification and industrial-grade systems.

8. Conclusion

This project successfully demonstrates the design and implementation of a *TTL-Based Aerospace Cabin Safety Logic Controller* using standard 7400-series TTL integrated circuits. The system processes different aircraft cabin operational conditions through combinational logic circuits and provides appropriate visual safety indications using LEDs. The controller successfully monitors pilot emergency conditions, passenger seat status, and lavatory occupancy conditions using input switches. Based on the implemented Boolean logic and truth table analysis, the circuit correctly generates safe, warning, and emergency output indications through green, yellow, and red LEDs.

The project highlights the practical application of fundamental digital electronics concepts including Boolean algebra, combinational logic design, truth table implementation, and TTL logic gate operation using simple and widely available hardware components. The system demonstrates how digital logic circuits can be applied to simulate aerospace-inspired monitoring and safety control systems without using any programmable device or microcontroller.

Although the project has some limitations such as fixed hardware logic and limited input conditions, it successfully achieves its educational objectives and provides a clear understanding of digital logic implementation and hardware-based safety monitoring systems. Overall, the project serves as an effective demonstration of practical TTL logic circuit design in a real-world inspired application.

9. GitHub Link

<https://github.com/amsami2120/ttl-aerospace-cabin-safety-controller>